

Microsoft releases Bing's search algorithm as an open source project on GitHub

Microsoft has open sourced an AI algorithm that enables its search engine Bing to deliver more relevant results to users, faster.

The Space Partition Tree And Graph (SPTAG) algorithm is one of the most advanced AI tools used by Microsoft to better meet people's evolving search needs. It allows users to take advantage of the intelligence from deep learning models to search through billions of pieces of information, called vectors, within milliseconds.

The AI algorithm has been released on the Microsoft-owned site, GitHub, as an open source project. This means the algorithm is now available to download, fork and tweak.

Vector search makes it easier to search by concept rather than keyword. According to Microsoft, it uses vector search for its own Bing search engine, and the technology is helping Bing better understand the intent behind billions of Web searches and find the most relevant result among billions of Web pages.

"Bing processes billions of documents every day, and the idea now is that we can represent these entries as vectors and search through this giant index of 100 billion-plus vectors to find the most related results in 5 milliseconds," said Jeffrey Zhu, programme manager on Microsoft's Bing team.

The Bing team said that it expects that the open source offering could be used for enterprise or consumer-facing applications to identify a language being spoken based on an audio snippet, or for image-heavy services.

"Even a couple of seconds for a search can make an app unusable," noted Rangan Majumder, group programme manager on Microsoft's Bing search and AI team. The team is also hoping that researchers and academics will use the algorithm to explore other areas of search breakthroughs. "We've only started to explore what's really possible around vector search at this depth," Majumder said.

The new solution allows IT organisations to easily adapt to the changing business demands along with cutting down IT operational expenses. The latest features of the solution are focused on containerised and cloud workload support, improved data protection capabilities and enhanced integration with the public cloud.

Commenting on the development, Brent Schroeder, CTO, Americas, for SUSE, said, "SUSE is powering digital transformation with agile open source solutions like SUSE Enterprise Storage that enable enterprises to continually innovate, compete and grow."



"We work with the community to develop innovative open source technology that we then test and harden to ensure it is enterprise ready. We aim to transform enterprise storage infrastructure with a truly open and unified, intelligent software-defined storage solution that reduces costs and alleviates proprietary hardware lock-in for customers. SUSE Enterprise Storage 6 reflects SUSE's ongoing commitment to making Ceph 'enterprise consumable'," said Schroeder.

Powered by Ceph Nautilus, SUSE Enterprise Storage 6 enables customers to seamlessly adapt to changing business demands by:

- Accelerating innovation as it further removes storage silos, giving customers easier access to different types of data, and enabling them to quickly extract information from data using cutting-edge search and analytics tools.
- Maximising application availability with faster and more granular backups that can now leverage low-cost public cloud resources for improved data protection.
- Responding to changing business demands faster with the ability to quickly and easily leverage public cloud resources as part of their storage infrastructure.

The solution also helps customers in reducing operational IT expenses by:

- Optimising data placement with the ability to automatically and efficiently move data between all tiers of storage based on policy, ensuring access to critical data when customers need it.
- Improving IT efficiency with a single, scalable storage solution that meets all storage requirements for containerised and cloud-enabled workloads, no matter where they are.

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